

Azure az-900 Preparations

✓ Core Skills Required for AZ-900 Certification

☁ 1. Cloud Concepts

- Understand **cloud computing models**: Public, Private, Hybrid
- Know **cloud benefits**: Scalability, Agility, Fault Tolerance, High Availability, Disaster Recovery
- Familiarity with **cloud service types**:
 - **IaaS** – Infrastructure as a Service
 - **PaaS** – Platform as a Service
 - **SaaS** – Software as a Service

🔗 2. Understanding of Core Azure Services

- Learn about **Azure core services** like:
 - **Compute**: Azure Virtual Machines, Azure Functions, App Services
 - **Storage**: Blob, Disk, Table, File
 - **Networking**: Virtual Network, VPN Gateway, Load Balancer, Application Gateway, CDN
 - **Databases**: Azure SQL, Cosmos DB
 - **Analytics**: Synapse, HDInsight, Azure Monitor

🔒 3. Azure Security, Privacy, Compliance, and Trust

- **Microsoft Entra ID (Azure AD)**: Identity and access management
- **RBAC** (Role-Based Access Control)
- **MFA (Multi-Factor Authentication)** and **SSO**
- **Azure Defender, Sentinel, Key Vault**
- **Compliance Offerings**: Microsoft Purview, Trust Center, and compliance manager

💰 4. Azure Pricing, SLA, and Lifecycle

- **Consumption-based model** (Pay-as-you-go, no upfront cost)
- **Total Cost of Ownership (TCO) Calculator** and **Pricing Calculator**
- **Azure SLAs**: 99.9%, 99.95%, 99.99% (depending on service)
- Understand **Service Lifecycle**: Preview, General Availability (GA), and Retirement

Core Characteristics of Cloud Computing (AZ-900)

- ◆ **Scalability** – Ability to increase or decrease resources to meet demand (vertically or horizontally).
- ◆ **Elasticity** – Automatic provisioning and de-provisioning of resources as workload changes, saving cost and optimizing performance.
- ◆ **Agility** – Quickly deploy and test new solutions, enabling faster innovation and go-to-market.
- ◆ **Fault Tolerance** – Systems are designed to continue operating smoothly even when some components fail.
- ◆ **Disaster Recovery** – Built-in backup and replication options ensure business continuity during major failures.
- ◆ **High Availability** – Redundant systems ensure uptime and accessibility with minimal interruption.
- ◆ **Pay-as-you-go** – Only pay for what you use, reducing capital expenses and improving cost efficiency.
- ◆ **Economies of Scale** – Cloud providers reduce costs through massive data center operations, passing savings to users.
- ◆ **Global Reach** – Access services and deploy applications across global regions and datacenters.
- ◆ **Security** – Built-in features like encryption, identity management, and compliance with global standards.
- ◆ **Resource Pooling** – Shared physical and virtual resources allocated dynamically to users based on demand.
- ◆ **Self-Service** – Users can provision and manage resources directly via portals or APIs without human interaction.

Essential Cloud Terms – AZ-900 Ready

Vertical Scaling (Scale Up)

Adding more power (CPU, RAM, etc.) to an existing machine.

Example: Upgrading a server from 8 GB RAM to 16 GB RAM.

Horizontal Scaling (Scale Out)

Adding more instances or machines to handle the load.

Example: Adding more servers behind a load balancer.

Economics of Scalability

Cloud providers optimize infrastructure at scale, reducing costs for customers. As usage increases, per-unit cost often decreases.

CapEx (Capital Expenditure)

Upfront investment in physical infrastructure (servers, hardware). Traditional IT model.

 *Buy once, depreciate over years.*

OpEx (Operational Expenditure)

Pay-as-you-go spending model for cloud services.

 *No upfront cost, billed monthly based on usage.*

Consumption-Based Model

Cloud billing model where you:

-  **No upfront cost**
-  **Pay only for what you use**
-  **No cost for unused resources**
-  **Stop paying when you stop using**

This helps **avoid overprovisioning**, manage costs effectively, and align IT spending with actual business demand.

Cloud Service Models – IaaS, PaaS, SaaS (AZ-900 Ready)

1. IaaS – Infrastructure as a Service

Provides virtualized computing resources over the internet — **servers, storage, networking**.

-  **You manage:** OS, apps, data
-  **Provider manages:** Hardware, virtualization, storage, networking

 **Example:** Microsoft Azure Virtual Machines, AWS EC2

 **Used when:**

- You want full control over OS and apps
- You're migrating legacy systems
- Building custom IT environments

2. PaaS – Platform as a Service

Provides a platform to develop, run, and manage applications without managing infrastructure.

-  **You manage:** Applications, data
-  **Provider manages:** OS, middleware, runtime, infrastructure

 **Example:** Azure App Services, Google App Engine, Heroku

 **Used when:**

- You're developing apps rapidly
- You want built-in scaling, patching, and deployment tools
- Avoid infrastructure complexity

3. SaaS – Software as a Service

Delivers ready-to-use software over the internet. You just use the app — **no infrastructure or platform management.**

-  **Provider manages everything**

 **Example:** Microsoft 365, Google Workspace, Salesforce, Dropbox

 **Used when:**

- You want ready-made apps for productivity, CRM, email, etc.
- You want low-maintenance, scalable, subscription-based tools

Cloud Deployment Models (AZ-900 Terminology)

Public Cloud

Cloud resources are owned and operated by a third-party cloud provider and shared with multiple customers over the internet.

Example: Microsoft Azure

Private Cloud

Cloud resources are used exclusively by one organization. It can be hosted on-premises or by a third party. Offers greater control and security.

Hybrid Cloud

Combines public and private clouds, allowing data and apps to move between them. Provides greater flexibility and more deployment options.

Multicloud

Use of **two or more** public cloud providers to avoid dependency on a single vendor. Each cloud may serve different workloads or services.

Azure Arc

A **Microsoft Azure service** that enables you to **manage on-premises, multi-cloud, and edge environments** using Azure management tools — bringing Azure services anywhere.

Azure Physical Infrastructure – Key Concepts (AZ-900)

◆ **Datacenters:** Physical buildings with racks of servers, power, cooling, and network equipment.

◆ **Region:** A geographical area containing one or more datacenters, connected via low-latency networks.
Example: Central India, East US.

◆ **Availability Zones:** Independent datacenters within a region with separate power, cooling, and networking for high availability and fault isolation.

◆ **Region Pair:** Two regions at least 300 miles apart, linked for disaster recovery and data residency. One is prioritized during outages or updates.

◆ **Sovereign Regions:** Special isolated Azure instances for specific compliance needs (e.g., Azure Government, Azure China).

Azure Management Infrastructure (AZ-900)

◆ *Resource*

Basic building block (VMs, databases, networks, etc.).

◆ *Resource Group*

Logical container to group related resources.

📌 Actions (like delete or permission) apply to all resources inside.

◆ *Subscription*

Unit for **billing**, **resource organization**, and **access control**.

Multiple subscriptions can exist under one account for isolation or cost tracking.

◆ *Management Group*

Used to **organize multiple subscriptions** into a hierarchy.

📌 Apply policies and access controls across all included subscriptions.

Azure Compute Services

💻 *Azure Virtual Machine (VM)*

- Emulates a **physical machine** in the cloud
- Offers **different hardware configurations** per VM
- Supports **multiple operating systems** (Windows, Linux, etc.)
- Full control over **file system, OS, and software**
- This is an **IaaS** offering (Infrastructure as a Service)

- You can deploy from **Azure Marketplace images** or **custom images**
✔ Best for: Full control, legacy apps, specific configurations

VM Scale Sets

- A group of **identical, load-balanced VMs**
- Enables **auto-scaling** based on demand
- Ensures **high availability** and consistent configuration
✔ Best for: Large-scale applications, microservices with VM-level control

Containers (e.g., Docker on Azure)

- Run **apps without their own OS**; they use the **host OS kernel**
- Lightweight, **faster to start**, and **require less compute/storage**
- Ideal for **microservices and DevOps pipelines**
- Use **images** to deploy apps instantly
✔ Best for: Fast deployment, portability, isolated environments

Azure Kubernetes Service (AKS)

- Fully managed **Kubernetes container orchestration**
- Automates **scaling, updates, and deployment** of containerized apps
- Supports **self-healing, load balancing, and rolling updates**
✔ Best for: Complex, scalable, microservice-based architectures

Azure Networking Services Explained (AZ-900)

Azure Virtual Network (VNet)

Think of a VNet as your **private cloud network**. It's like setting up your own company network in the cloud—where your virtual machines and other services can securely talk to each other.

Network Security Group (NSG)

An NSG acts like a **firewall**. It lets you control **what traffic is allowed in or out** of your Azure resources like VMs or subnets.

You define **rules** based on IP, port, or protocol.

Example:

- Allow port 80 (HTTP)
- Deny everything else

VPN Gateway

Used to **connect your on-premises network to Azure** over a secure, **encrypted VPN tunnel**.

It's like running a private cable from your office to Azure—but using the internet securely.

 **Use case:** Secure hybrid cloud setups.

Azure Load Balancer

Distributes incoming traffic to **multiple VMs** to keep your app highly available.

It works at **Layer 4 (Transport Layer)** of the OSI model.

 **Think of it as:** Traffic police that balances traffic to prevent server overload.

Azure Application Gateway

An advanced load balancer for **web applications** that works at **Layer 7 (Application Layer)**.

It can do things like **URL-based routing**, **SSL termination**, and **Web Application Firewall (WAF)** integration.

Example:

- Requests to /login go to Server A
- Requests to /checkout go to Server B

Content Delivery Network (CDN)

Delivers **web content like images, videos, or scripts** from servers closest to the user.
Improves **performance** and **reduces latency** by caching content globally.

 **Think of it as:** A worldwide system of mirrors for fast content delivery.

Azure Storage Services – AZ-900 Guide

Azure Blob Storage

Blob = Binary Large Object

Used for **unstructured data** like images, videos, backups, logs, documents.

Use cases:

- Storing media files, logs
- Hosting static website files
- Backup & restore, disaster recovery

Access Tiers:

- **Hot:** Frequent access (e.g., daily user files)
- **Cool:** Infrequent access (e.g., backups older than 30 days)
- **Archive:** Rarely accessed (e.g., long-term compliance data) – lowest cost, high retrieval time

 **Keyword:** Object storage | Tiered storage | Cost-effective archiving

2 Azure Queue Storage

A **message queuing system** used for reliable communication between different application components.

Use cases:

- Decoupling front-end and back-end processes
- Building scalable, asynchronous systems
- Processing task queues (e.g., order processing)

 **Keyword:** Asynchronous communication | Queue-based messaging | Durable messaging

3 Azure Table Storage

A **NoSQL key-value store** for structured, non-relational data.

It stores **massive amounts of semi-structured data** at low cost.

Use cases:

- Storing user data, settings, device logs
- When you need fast lookup with a Partition + Row Key

 **Keyword:** NoSQL | Key-value store | Structured data | Scalable

4 Azure File Storage

A **fully managed file share** in the cloud that can be **mounted via SMB protocol** (like a network drive).

Use cases:

- Lift-and-shift applications using file shares
- File sharing across distributed applications
- Mounting shared drives in Windows/Linux VMs

 **Keyword:** Server Message Block (SMB) | Shared file system | Cloud-mounted drive

Azure Disk Storage

Provides **persistent block-level storage** for Azure VMs.

Think of it like the **hard drive** for your virtual machine.

Use cases:

- Boot and data volumes for VMs
- High-performance workloads like databases or SAP
- Durable, secure storage with backups and encryption

Types:

- **Standard HDD** – Low-cost dev/test
- **Standard SSD** – Better latency
- **Premium SSD** – High performance
- **Ultra Disk** – Ultra-low latency, high IOPS

 **Keyword:** Managed Disks | VM Storage | Block-level | High availability

Microsoft Entra Overview (formerly Azure Active Directory)

1. Microsoft Entra ID (previously Azure AD)

Microsoft Entra ID is a **cloud-based identity and access management (IAM) service**. It helps organizations **manage users, groups, and permissions** securely.

Who uses it?

- **Organizations** using Microsoft 365 or Azure automatically use Microsoft Entra ID.
- **Employees** use it to access apps like Outlook, Teams, SharePoint, and custom enterprise apps.

Key Features:

- **SSO (Single Sign-On):** Sign in once and access multiple apps (e.g., M365, Salesforce, etc.).
- **Application Management:** Manage access to 1000s of SaaS apps.

- **Device Management:** Control which devices can access your organization's data.

2. Microsoft Entra Connect

- A tool used to sync your on-premises **Active Directory (AD)** with **Microsoft Entra ID**.
-  Ensures **hybrid identity** so users can use the **same credentials** on-prem and in Azure.

3. Microsoft Entra Domain Services

- Provides **domain join, group policy, and LDAP/NTLM authentication** in Azure—**without needing on-prem AD controllers**.
- Used when you need **legacy compatibility** for VMs and apps in the cloud.



Azure Authentication Methods

1. SSO – Single Sign-On

- Sign in once to access all company apps securely.
- Example: Log in once to Microsoft 365 and access Teams, Outlook, SharePoint without re-authenticating.

2. MFA – Multi-Factor Authentication

- Requires **2+ authentication methods**:
 - Something you know (password)
 - Something you have (phone)
 - Something you are (biometric)

Example: Logging in with a password + receiving a code on your phone.

3. Password-less Authentication

- Sign in **without a password**, using secure alternatives:

- **Windows Hello for Business** (Face/Pin)
- **FIDO2 Security Keys**
- **Microsoft Authenticator App** (tap to approve)

Conditional Access (smart security control)

A **policy-based system** that decides **when and how users can access resources**, based on:

- **Signals:** User, location, device, app
- **Decisions:** Allow, Block, Require MFA
- **Enforcement:** Controls applied (e.g., block access if outside allowed location)

Example: Require MFA only if login is from a new country or unfamiliar device.

RBAC – Role-Based Access Control

- Controls who has access to **what resources** in Azure.
- Access is granted based on the **user's role** (e.g., Reader, Contributor, Owner).

Example: A developer can manage VMs but not delete storage accounts.

Zero Trust Security Model

Core principles:

1. **Verify Explicitly** – Always authenticate and authorize.
2. **Use Least Privilege Access** – Give minimum permissions needed.
3. **Assume Breach** – Always prepare for threats even from inside the network.

 **Zero Trust uses:**

- Conditional Access

- MFA
- Just-in-time (JIT) access
- Segmented networks
- Risk-based policies

Defense in Depth (Layered Security)

A **multi-layered security strategy** where each layer protects against threats.

Layers:

1. **Data** – Encryption, access controls
2. **Application** – Secure code, API protection
3. **Compute** – Patch VMs, endpoint protection
4. **Networking** – Firewalls, NSGs, DDoS protection
5. **Perimeter** – Threat detection, IDS/IPS
6. **Identity & Access** – MFA, SSO, RBAC
7. **Physical** – Secure datacenters (biometric, guards, etc.)

Microsoft Defender for Cloud

- A **Cloud-Native Application Protection Platform (CNAPP)**.
- Helps you **assess**, **secure**, and **defend** your Azure (and multi-cloud) environment.

What it does:

- Assesses security posture (Secure Score)
- Detects threats (e.g., brute force, malware)
- Recommends fixes
- Works across Azure, AWS, and GCP



1. Azure Cost Management



Factors That Affect Azure Costs

- **Resource type:** VMs, storage, bandwidth, etc.
- **Location/Region:** Prices vary by Azure region.
- **Usage:** Pay-as-you-go based on consumption.
- **Reserved Instances:** Discounted pricing for long-term commitments.
- **Licensing:** Software licenses (e.g., Windows Server) may affect cost.
- **Network Traffic:** Inbound is free; outbound may be charged.



Pricing Calculator vs. TCO Calculator

Tool	Purpose
Pricing Calculator	Estimates cost of specific services before deployment.
TCO Calculator	Compares on-premises infrastructure cost vs. Azure, including hardware, electricity, and IT labor savings.



Cost Management in Azure

Azure Cost Management helps:

- Analyze spending patterns
- Set budgets and alerts
- Optimize underused resources
- Forecast future usage



Tags in Azure

Tags = name/value pairs to **organize resources** by:

- Department
- Cost center
- Environment (e.g., dev, prod)

Helps in **cost tracking, automation, and governance**.

2. Azure Governance & Compliance Tools

Microsoft Purview

- Unified **data governance** tool.
- Helps track, label, classify, and **manage sensitive data** across Azure, Microsoft 365, and on-prem.

Azure Policy

- Enforces organizational rules on resources.
- Examples:
 - Allow only specific VM sizes
 - Enforce tags
 - Disallow creation outside certain regions

Resource Locks

Prevent accidental deletion or changes:

- **ReadOnly** – can view but not change
- **Delete** – can change but not delete



3. Azure Resource Management & Deployment



Azure Portal

- Web-based UI for managing resources.



Azure Cloud Shell

- Browser-based shell environment.
- Includes:
 - **Azure CLI** (command-line for scripts)
 - **Azure PowerShell** (cmdlets for resource automation)



Azure Arc

- Manage **on-prem, edge, and multi-cloud resources** from Azure.
- Extend Azure services like Policy or Monitor to non-Azure servers.



Infrastructure as Code (IaC)

- Use **code to deploy/manage infrastructure**, not manual steps.
- Improves **consistency, repeatability, and automation**.



Azure Resource Manager (ARM) & Templates

- ARM = Azure's deployment and management engine.
- **ARM templates** = JSON files to define infrastructure as code.
- Deploy entire environments repeatedly and predictably.

4. Azure Monitoring Tools

Azure Advisor

- Personalized best practices for:
 - Cost savings
 - High availability
 - Performance
 - Security

Azure Service Health

- Real-time alerts about:
 - Outages
 - Planned maintenance
 - Health advisories specific to your services

Azure Monitor

Central hub for telemetry data:

- **Log Analytics** – Query logs from apps, VMs, and services.
- **Alerts** – Notifications when metrics cross thresholds.
- **Application Insights** – Monitor app performance and errors.

26 AZ-900 Exam Keywords Explained

1. **Azure Resource Manager Template (ARM Template)**

Not just for consistency—these JSON-based templates allow automated and repeatable deployments of Azure resources across dev, test, and production environments.

2. **Firewall**

A security service that controls inbound and outbound traffic across multiple virtual networks, helping protect Azure resources like VMs and subnets.

3. **VPN Gateway**

A secure tunnel that encrypts traffic between your on-premises network and Azure—ideal for hybrid cloud setups.

4. **ExpressRoute**

Creates a private connection (not over public internet) between your infrastructure and Microsoft data centers via a connectivity provider.

5. **Infrastructure as Code (IaC) Principle: Idempotency**

No matter how many times you run a deployment, the result is the same. This eliminates configuration drift and enhances automation.

6. **GitLab**

A DevOps tool that helps store code, track changes, collaborate with teams, and manage version control using Git.

7. **GitHub Actions**

Automate workflows like build, test, and deploy. Supports Continuous Integration and Continuous Deployment (CI/CD) pipelines.

8. **Azure DevOps**

A suite of tools supporting the software development lifecycle: Azure Boards (tracking), Repos (Git), Pipelines (CI/CD), Artifacts (package mgmt), and Test Plans.

9. **Azure DevTest Labs**

Lets developers create pre-configured VMs for test environments without investing in physical hardware—great for saving cost and time.

10. **HDInsight**

A cloud-based PaaS service for processing massive amounts of data using Apache Hadoop, Spark, and Kafka. Supports real-time data streams and ML jobs.

11. **IoT Hub**

A PaaS messaging hub for managing millions of IoT devices. Enables secure, two-way communication between IoT devices and the cloud.

12. **IoT Central**

A SaaS IoT solution with a prebuilt UI. Best for teams with less development effort. Manages devices and analytics through a simple dashboard.

13. **Azure Sphere**

A comprehensive IoT security solution combining certified hardware, OS, and cloud-based security for building secured IoT devices.

14. **Cosmos DB**

A globally distributed, fully managed NoSQL database. Great for low-latency, real-time apps at a global scale. Supports auto-scaling and multi-model APIs.

15. **Azure Database for PostgreSQL**

A fully managed, open-source relational database service. Ideal for users migrating their PostgreSQL workloads to Azure with built-in scalability.

16. **Azure Synapse Analytics**

A powerful analytics service and enterprise data warehouse that combines data integration, warehousing, and big data analytics.

17. **Azure CLI**

A cross-platform command-line tool to create, manage, and automate Azure resources using simple commands.

18. **Azure Portal**

A web-based GUI that provides full access to manage Azure resources. A user-friendly alternative to command-line tools.

19. **Azure PowerShell**

A scripting environment using cmdlets to manage Azure resources, ideal for automating repetitive tasks.

20. **PowerShell (CMDLets)**

Small command units in scripting, used to automate tasks and manage Azure environments.

21. **Cloud Shell**

An in-browser command-line interface with pre-installed tools like Azure CLI and PowerShell. Accessible from the Azure Portal.

22. **Elasticity**

The system's ability to auto-scale resources (like VMs or services) up or down instantly depending on demand—reduces cost and waste.

23. **Agility**

The ability of an organization to quickly develop, test, and deploy solutions to adapt to changes in market or business needs.

24. **Fault Tolerance**

The system's ability to continue operating even when some components fail. Helps maintain uptime and service continuity.

25. **SLA Uptime**

Azure guarantees 99.99% uptime for VMs deployed in an Availability Set or Zone. Azure Kubernetes Service (AKS) has a 99.5% SLA.

26. **Microsoft Defender for Cloud**

A built-in security service in Azure that helps monitor, assess, and protect resources from threats. Offers Just-In-Time (JIT) access for VMs and alerts on suspicious activity.

27. **Azure Management Methods**

Manage Azure using:

- **Azure Portal:** Web UI to manage resources
- **PowerShell:** Uses cmdlets, script automation on Windows

- **Azure CLI:** Command-line tool, great for Linux
- **Cloud Shell:** Web-based shell that supports both PowerShell and Bash
- **Azure Region vs Availability Zone**

An Azure Region may or may not have Availability Zones. A region has one or more datacenters with power, cooling, and networking infrastructure.

28. Data Lake

Storage system for massive amounts of structured and unstructured data. Ideal for big data analytics and long-term storage. Works with IoT data and other sources.

29. Containers

Lightweight and portable app packages. No OS of their own—they share the host OS. Fast, efficient, and serverless.

30. Logic Apps

No-code/low-code automation tool. Automate workflows in apps like Outlook (e.g., auto-save attachments, send alerts). Serverless and ideal for non-developers.

31. Kubernetes

Container orchestration tool. Manages scaling, networking, and restarts of multiple containers. Automates deployment and operations.

32. Azure Functions

Run small pieces of code on demand (trigger-based). Triggers can be uploads, HTTP requests, timers, etc. Serverless and event-driven.

33. App Service

PaaS offering to host web and mobile apps without managing infrastructure. Supports multiple programming languages.

34. Microsoft Purview Compliance Portal

Provides detailed insights into compliance, security, and privacy features across Microsoft Cloud. Useful for governance and regulatory needs.

35. Data Retention in Blob Storage Tiers

- **Hot:** No minimum retention
- **Cold:** 30-day minimum
- **Archive:** 180-day minimum

36. Azure Firewall vs NSG

- **Azure Firewall:** Centralized filtering across subscriptions and VNets
- **Network Security Group (NSG):** Granular filtering at subnet and NIC level

37. DDoS Protection

Protects against large-scale malicious attacks on your apps/resources from multiple sources. Works at the network layer.

38. Azure Firewall

Stateful security service for network-wide traffic filtering. Centralized rule management with application and network rules.

39. Network Security Group (NSG)

Filters traffic by IP, port, and protocol at subnet or NIC level. Stateless rules for fine-grained control.

40. Azure Synapse Analytics

End-to-end analytics service that brings together big data and data warehousing. Analyzes data from lakes, databases, and external systems.

41. Azure AD Connect

Connects on-prem Active Directory to Azure Entra ID, enabling single sign-on and unified credentials.

42. Azure Key Vault

Securely stores secrets, certificates, and keys. Helps apps access databases or services without exposing sensitive credentials.

43. Azure Sentinel

SIEM tool for detecting, analyzing, and responding to security threats using AI. Monitors logs, detects anomalies, and helps incident response.

44. Azure Monitor

Tracks performance of Azure resources. Generates alerts and enables automated actions like scaling or rebooting.

45. Azure Advisor

Gives recommendations to improve cost, performance, security, and high availability of Azure resources.

46. Microsoft Purview (Data Governance)

Like a smart librarian + security guard. Organizes data across platforms and ensures governance, security, and compliance.